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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**

3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SSH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SSH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtmpadr proto kernel_r
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:                0.0
Usage of /:                  5.1% of 48.91GB
Memory usage:               5%
Swap usage:                 0%
Processes:                  139
Users logged in:            1
IPv4 address for enp0s3:    192.168.1.14
IPv6 address for enp0s3:    2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

Name: Genesis De Raya

Student ID: 20120206

Course/Year/Section: BSIT 3B

Date Submitted: _____

Instructor: _____

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ☐ Create and configure a Virtual Machine with Ubuntu Server
- ☐ Configure Bridged Network Adapter for network access
- ☐ Determine and verify the server's IP address
- ☐ Install OpenSSH Server and enable the service
- ☐ Establish successful SSH connection from host machine
- ☐ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Ubuntu
Virtualization Software (VirtualBox/VMware)	Oracle VM VirtualBox 7.0
Ubuntu Server Version	Ubuntu server 26.04LTS
VM Name	DeRayaServer
RAM Allocated	5549MB
Storage Allocated	100gb
Network Mode	Host-only Adapter
SSH Client Used	Secure shell client

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	5549MB
Hard Disk	100GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION

Commands Executed:

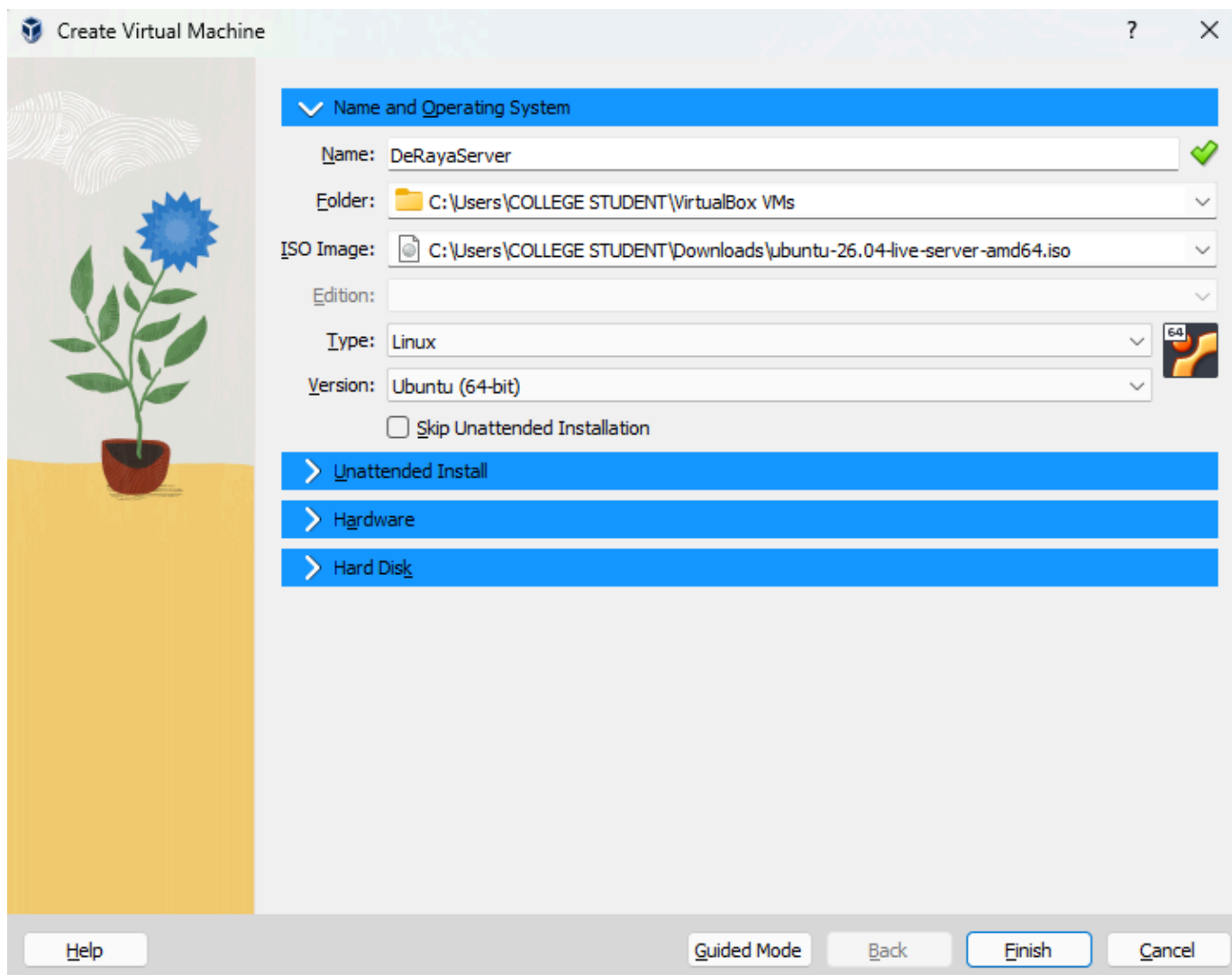
Server IP Address: 192.168.56.101

Network Interface: Bridged network

6. REMOTE CONNECTION

Connection Details:

Parameter	Value
Server IP	192.168.56.101
Username	chobi
SSH Port	ssh chobi@ 192.168.56.101



Create Virtual Machine

✓ Name and Operating System

Name: DeRayaServer ✓

Folder: C:\Users\COLLEGE STUDENT\VirtualBox VMs

ISO Image: C:\Users\COLLEGE STUDENT\Downloads\ubuntu-26.04-live-server-amd64.iso

Edition:

Type: Linux

Version: Ubuntu (64-bit)

☐ Skip Unattended Installation

> Unattended Install

> Hardware

> Hard Disk

Help Guided Mode Back Finish Cancel

Create Virtual Machine

> Name and Operating System

> Unattended Install

Username and Password

Username: vboxuser

Password:

Repeat Password:

Additional Options

Product Key: #####-#####-#####-###

Hostname: DeRayaServer

Domain Name: myguest.virtualbox.org

☐ Install in Background

☐ Guest Additions

Guest Additions ISO: <not selected>

> Hardware

> Hard Disk

Help

Guided Mode

Back

Finish

Cancel

Create Virtual Machine

> Name and Operating System

> Unattended Install

> Hardware

Base Memory: 5549 MB

4 MB 16384 MB

Processors: 4

1 CPU 12 CPUs

☐ Enable EFI (special OSes only)

> Hard Disk

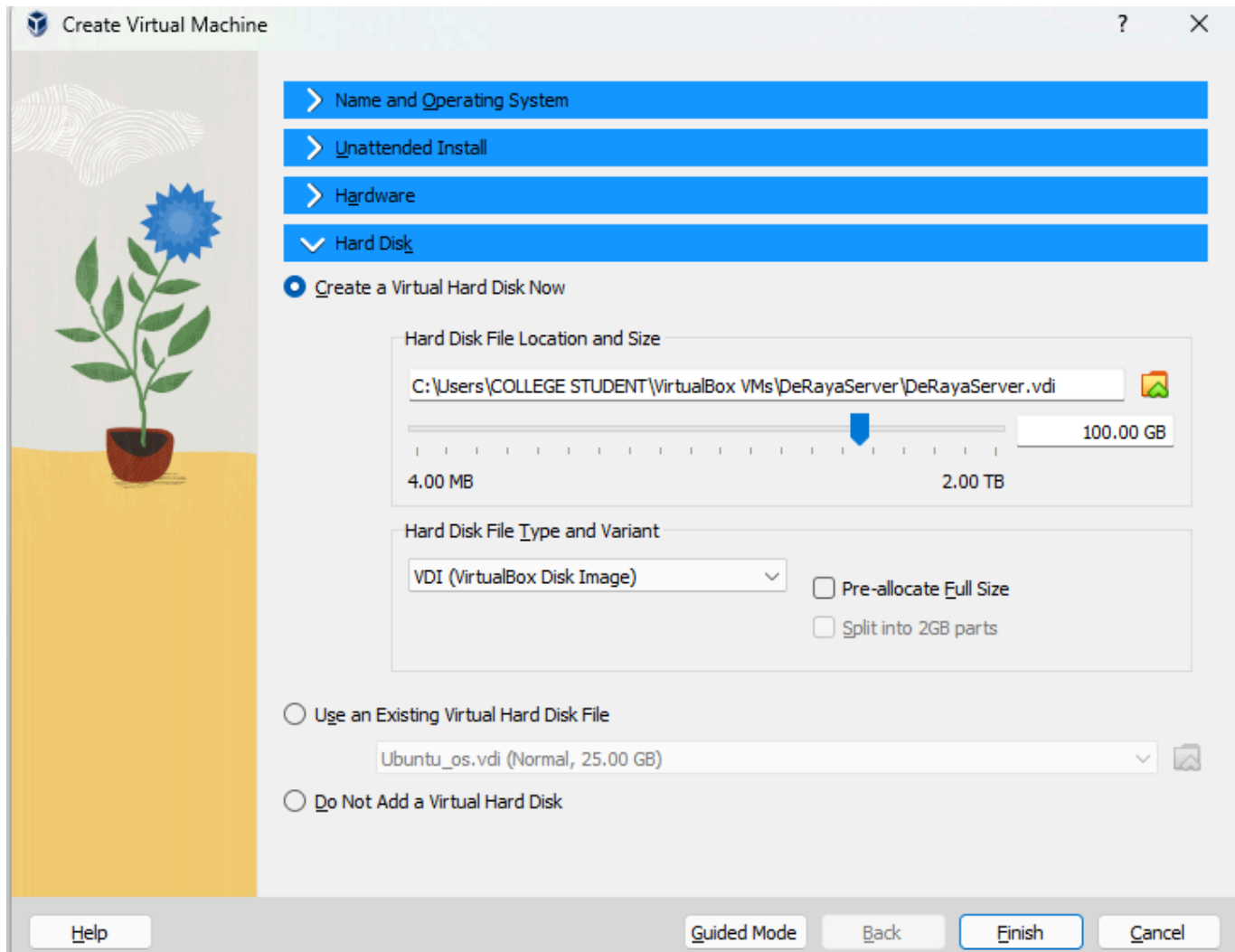
Help

Guided Mode

Back

Finish

Cancel



Installer update available

Version 26.04.1 of the installer is now available (26.04 is currently running).

You can read the release notes for each version at:

<https://github.com/canonical/subiquity/releases>

If you choose to update, the update will be downloaded and the installation will continue from here.

[Update to the new installer]
[Continue without updating]
[Back]

Choose the type of installation

Choose the base for the installation.

(X) Ubuntu Server

The default install contains a curated set of packages that provide a comfortable experience for operating your

() Ubuntu Server (minimized)

This version has been customized to have a small runtime footprint in environments where humans are not expected in.

Additional options

[] Search for third-party drivers

This software is subject to license terms included with its documentation. Some is proprietary. Third-party drivers should not be installed on systems that will be used for FIPS or the real-time kernel.

[Done]
[Back]

Network configuration

Configure at least one interface this server can use to talk to other machines, and which preferably provides sufficient access for updates.

NAME	TYPE	NOTES
[enp0s3	eth	- ▶]
DHCPv4	10.0.2.15/24	
DHCPv6	-	
08:00:27:22:b6:80 / Intel Corporation / 82540EM Gigabit Ethernet Controller (PRO/1000 MT Desktop Adapter)		

[Create bond ▶]

[Done]
[Back]

Proxy configuration

If this system requires a proxy to connect to the internet, enter its details here.

Proxy address:

If you need to use a HTTP proxy to access the outside world, enter the proxy information here. Otherwise leave this blank.

The proxy information should be given in the standard form of "http://[user][:pass]@host[:port]/".

[Done]
[Back]

Ubuntu archive mirror configuration

If you use an alternative mirror for Ubuntu, enter its details here.

Mirror address: `http://ph.archive.ubuntu.com/ubuntu/`

You may provide an archive mirror to be used instead of the default.

This mirror location passed tests.

```
Ign:1 file:/cdrom resolute InRelease
Get:2 file:/cdrom resolute Release [664 B]
Get:2 file:/cdrom resolute Release [664 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:4 http://ph.archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:5 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:6 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Fetched 409 kB in 0s (856 kB/s)
Reading package lists...
```

[Done]
[Back]

Guided storage configuration

Configure a guided storage layout, or create a custom one:

(X) Use an entire disk

[VBOX_HARDDISK_VB9c4d6c65-2d58cc40 local disk 100.000G ▼]

[X] Set up this disk as an LVM group

[] Encrypt the LVM group with LUKS

Passphrase:

Confirm passphrase:

[] Also create a recovery key
The key will be stored as ~/recovery-key.txt in the live system and will be copied to /var/log/installer/ in the target system.

() Custom storage layout

[Done]
[Back]

Storage configuration

FILE SYSTEM SUMMARY

MOUNT POINT	SIZE	TYPE	DEVICE TYPE
[/	48.996G	new ext4	new LVM logical volume ▶]
[/boot	2.000G	new ext4	new partition of local disk ▶]

AVAILABLE DEVICES

DEVICE	TYPE	SIZE
[ubuntu-vg (new)	LVM volume group	97.996G ▶]
free space		49.000G ▶]

[Create software RAID (md) ▶]
[Create volume group (LVM) ▶]

USED DEVICES

DEVICE	TYPE	SIZE
[ubuntu-vg (new)	LVM volume group	97.996G ▶]
ubuntu-lv new, to be formatted as ext4, mounted at /		48.996G ▶]

[VBOX_HARDDISK_VB9c4d6c65-2d58cc40 local disk 100.000G ▶]
partition 1 new, BIOS grub spacer 1.000M ▶]
partition 2 new, to be formatted as ext4, mounted at /boot 2.000G ▶]
partition 3 new, PV of LVM volume group ubuntu-vg 97.997G ▶]

[Done]
[Reset]
[Back]

Storage configuration

FILE SYSTEM SUMMARY

MOUNT POINT	SIZE	TYPE	DEVICE TYPE
[/	48.996G	new ext4	new LVM logical volume ▶]
[/boot	2.000G	new ext4	new partition of local disk ▶]

AVAILABLE DEVICES

DEVICE	TYPE	SIZE
[ubuntu-vg (new)	LVM volume group	97.996G ▶]
free space		49.000G ▶]

[Create software RAID (md) ▶]
[Create volume group (LVM) ▶]

USED DEVICES

DEVICE	TYPE	SIZE
[ubuntu-vg (new)	LVM volume group	97.996G ▶]
ubuntu-lv new, to		48.996G ▶]

[VBOX_HARDDISK_VB9c4d6 partition 1 new, BIO
partition 2 new, to
partition 3 new, PV

Confirm destructive action

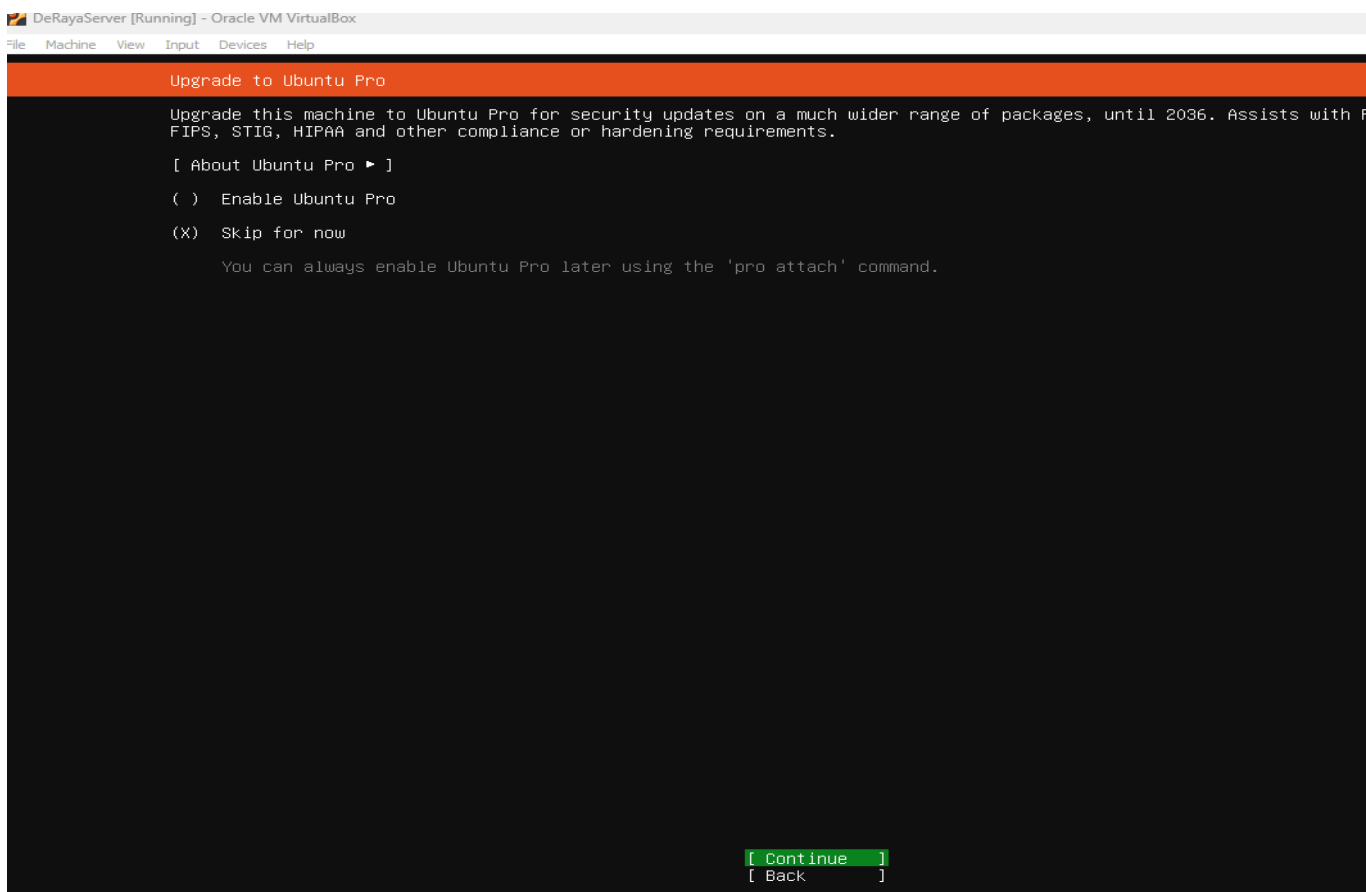
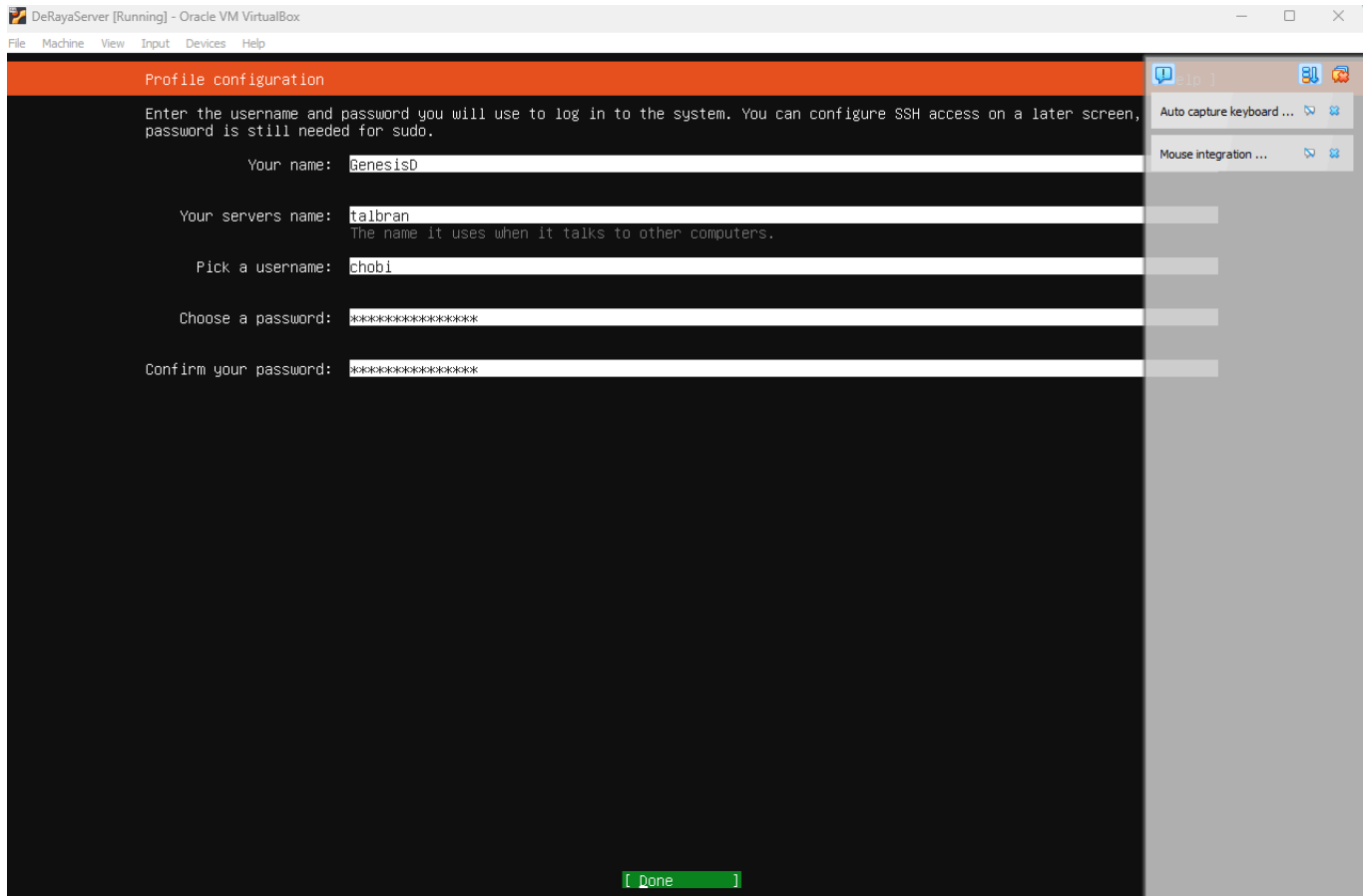
Selecting Continue below will begin the installation process and result in the loss of data on the disks selected to be formatted.

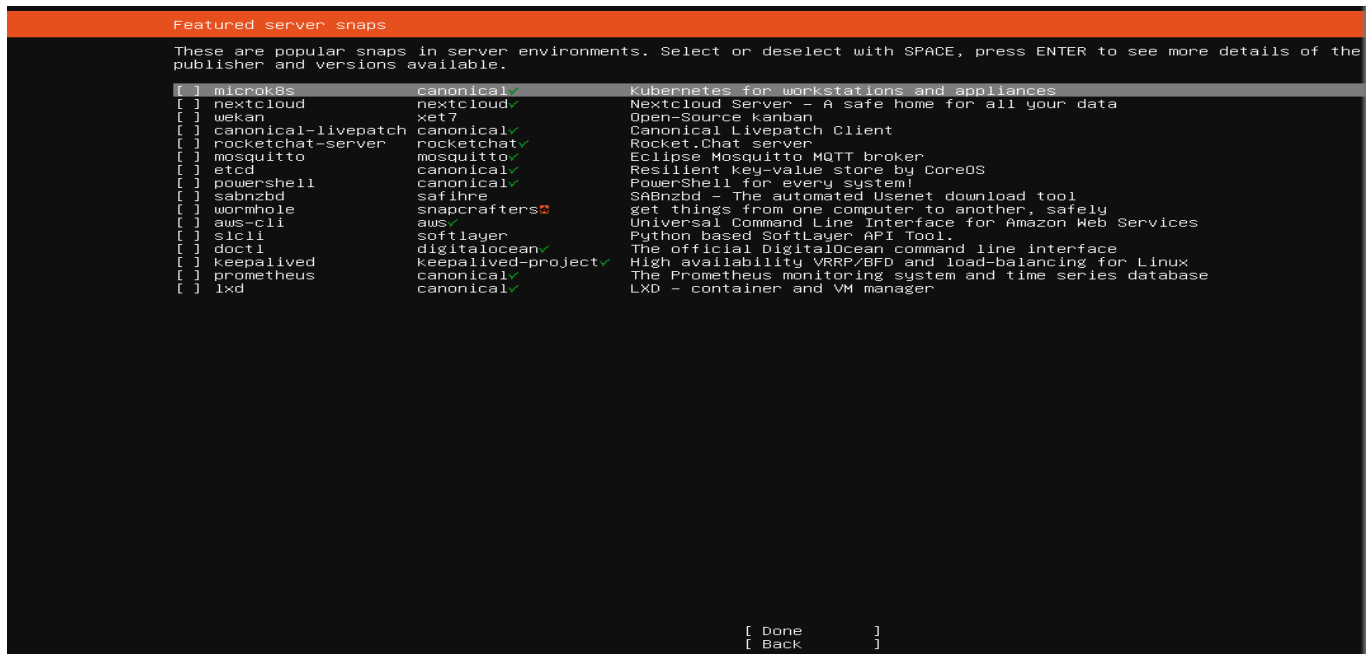
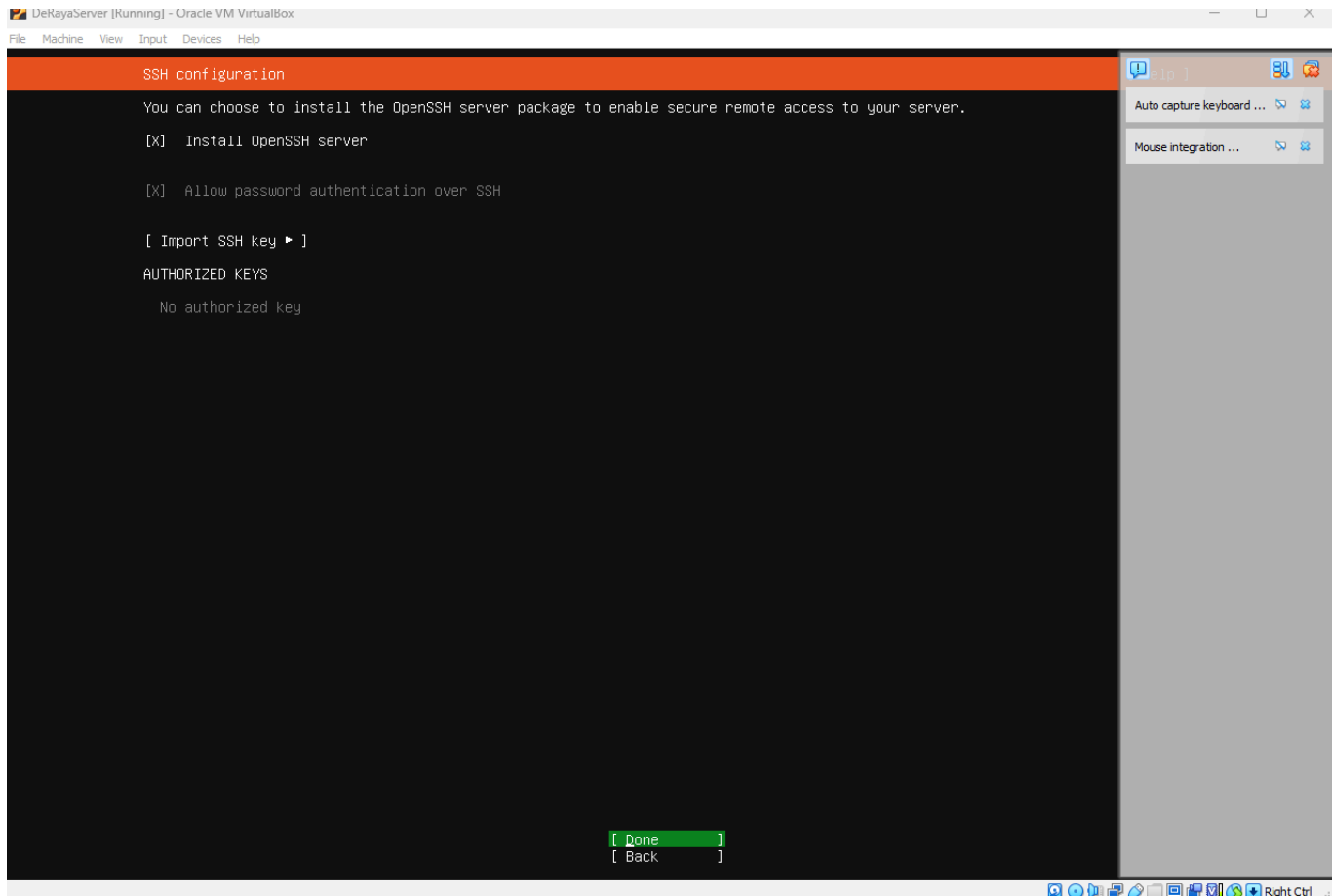
You will not be able to return to this or a previous screen once the installation has started.

Are you sure you want to continue?

[No]
[Continue]

[Done]
[Reset]
[Back]





DeRayaserver [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Updating system

```
configuring mount: mount=1
configuring mount: mount=0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpput8xq12/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
retrieving openssh-server
curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target
```

[View full log]

Auto capture keyboard ...

Mouse integration ...

DeRayaserver [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

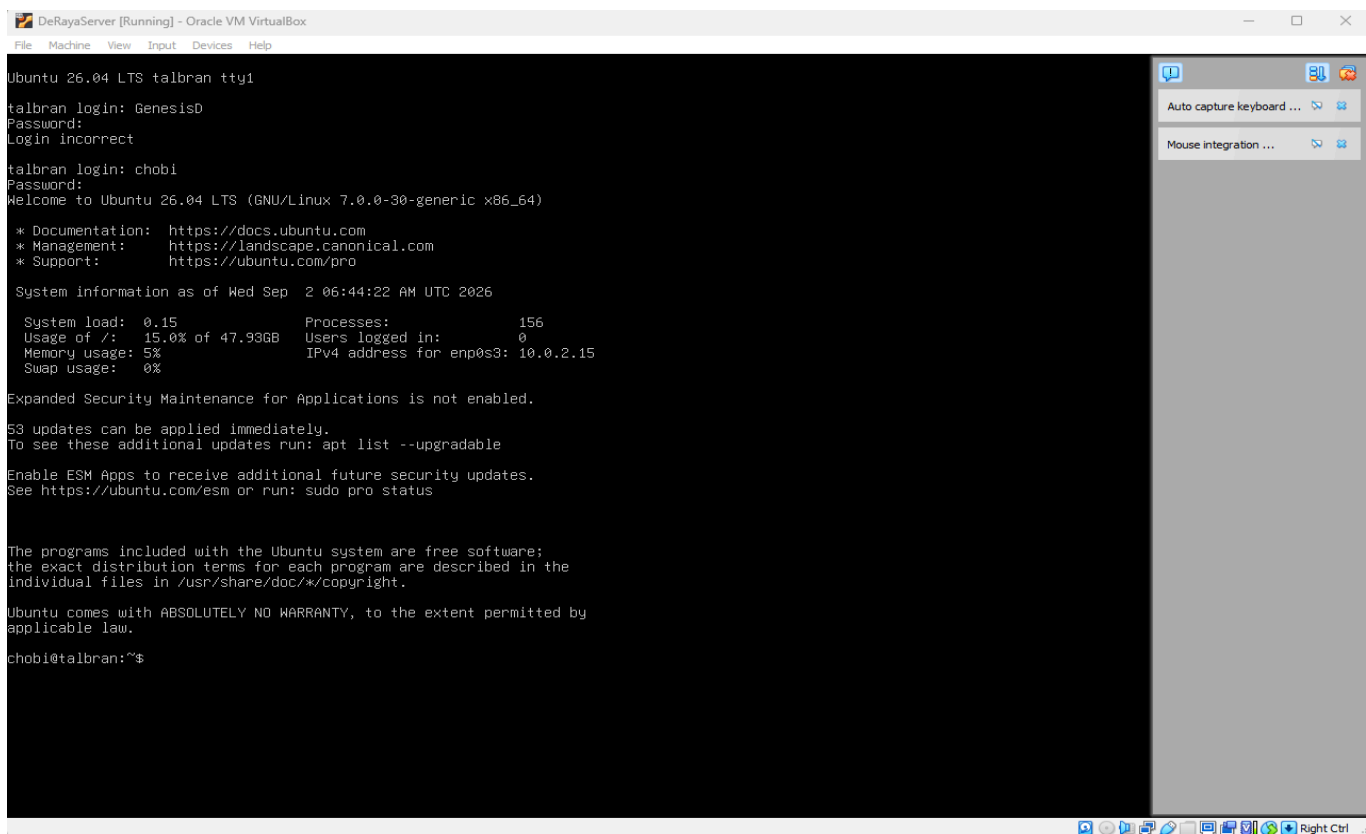
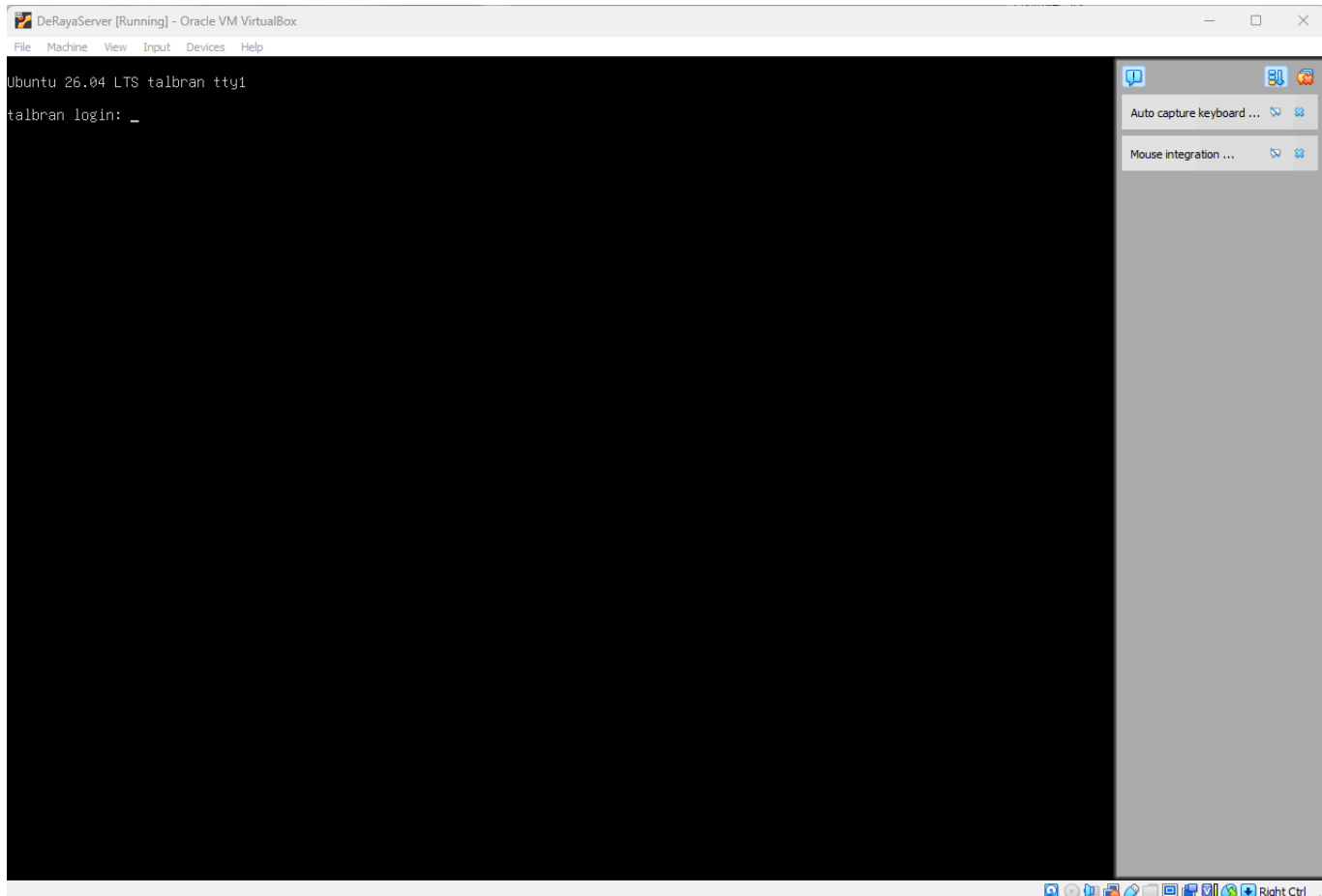
Installation complete!

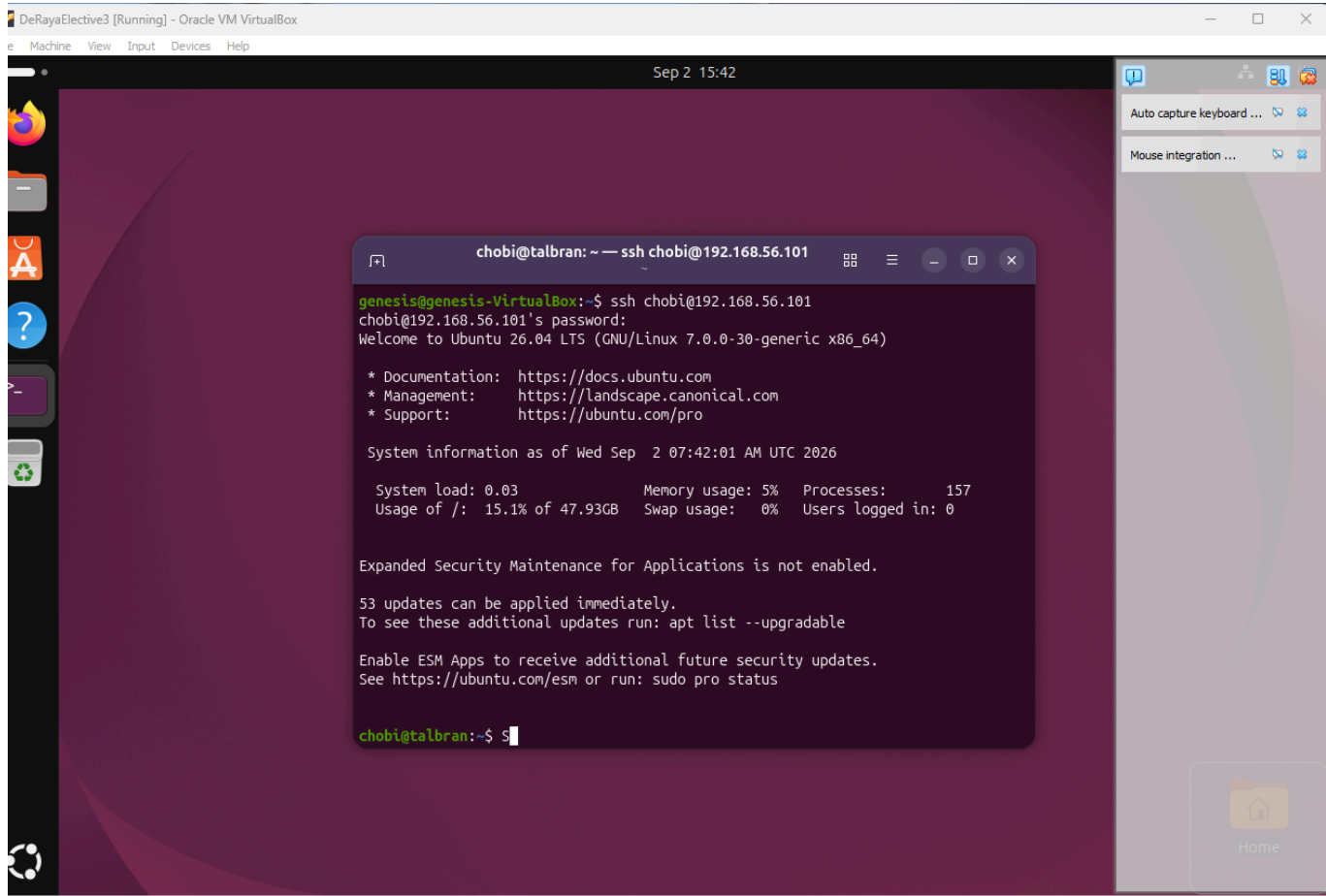
[Help]

```
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpput8xq12/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
retrieving openssh-server
curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/late/run_builtin:
subiquity/late/run_user_supplied:
```

[View full log]

[Reboot Now]






```
DeRayaserver [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Ubuntu 26.04 LTS talbran tty1
talbran login: chobi
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep  2 07:40:27 AM UTC 2026

System load: 0.0      Memory usage: 5%    Processes:   153
Usage of /:  15.1% of 47.93GB  Swap usage:  0%    Users logged in: 0

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

chobi@talbran:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:22:b6:80 brd ff:ff:ff:ff:ff:ff
    altname enx08002722b680
    inet 192.168.56.101/24 metric 100 brd 192.168.56.255 scope global dynamic enp0s3
        valid_lft 441sec preferred_lft 441sec
    inet6 fe80::a00:27ff:fe22:b680/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
chobi@talbran:~$ whoami
chobi
chobi@talbran:~$
```